Install for a local PC

0. Preparation

Before installing srcroot make sure that the following packages are installed on your system

- Install packages needed for RedHat-based OS (eg, CentOS, Scientific Linux, AlmaLinux):

```
sudo su

yum install -y --skip-broken epel-release && yum install -y --skip-broken
libuuid-devel wget rsync subversion git make cmake gcc-gfortran gcc-c++
binutils patch redhat-lsb-core redhat-lsb libbsd-devel libicu-devel
libX11-devel libXmu-devel libXpm-devel libXft-devel libXext-devel
mesalibGLU-devel ncurses-devel python3-devel libxml2-devel expat-devel
zlibdevel postgresql-devel mysql mysql-devel sqlite-devel openssl-devel
curldevel automake libtool readline-devel xz-devel xerces-c-devel
protobufdevel gsl-devel fftw3-devel && yum --enablerepo=crb install -y
protobufdevel mysql-devel &> /dev/null
```

- Install packages needed for Debian-based OS (eg, Ubuntu):

```
sudo su

apt-get -m -y install uuid-dev wget rsync subversion git make cmake g++
gcc gfortran binutils patch lsb-release libicu-dev libx11-dev libxmu-dev
libxpm-dev libxft-dev libxext-dev dpkg-dev xlibmesa-glu-dev libglew-dev
python3-dev python-dev-is-python3 libxml2-dev libexpat1-dev zlib1g-dev
libpqxx-dev libmysqlclient-dev default-libmysqlclient-dev libsqlite3-dev
libssl-dev libcurl4-openssl-dev automake libtool libreadline-dev liblzma-
dev libxerces-c-dev libgsl-dev libfftw3-dev protobuf-compiler-grpc &&
apt-get -m -y install libcrypt-dev &> /dev/null
```

Note that the current software packages requires GCC 7.2.1+, CMake 3.16.1+ and PostgreSQL client 10+ (you can check it by `gcc -v`, `cmake --version` and `psql -version` commands).

If your operating system does not meet the requirements, install the latest version of GCC, CMake, PostgreSQL client or auxiliary devtoolset and cmake3 packages. For instance, for CentOS 7 run: `sudo yum install -y centos-release-scl cmake3; sudo yum install -y devtoolset-9`. To enable the devtoolset, run `scl enable devtoolset-9 -- bash` (every time to use it). To employ cmake3 package, just use `cmake3` command instead of the usual cmake command. To install new PostgreSQL client versions. you can run (e.g. for version 14) `sudo yum install -y` https://download.postgresql.org/pub/repos/yum/reporpms/EL-7-x86_64/pgdgredhat-repo-latest.noarch.rpm; sudo yum install -y postgresql14-devel. To update cmake package for an old Ubuntu/Debian system, you can use kitware repository.

Installing the FairSoft and FairRoot packages

1.Set an installation path for the external packages, e.g. /opt
NOTE: If you are installing FairSoft/FairRoot to a system directory like /opt, then switch to superuser before (**"sudo su" command**).

```
export INSTALLATION_PATH=/opt

cd $INSTALLATION_PATH
```

2.Install the FairSoft package
The February 2026 release of FairSoft (with ROOT 6.36.08) can be downloaded from GitHub:

```
git clone https://github.com/FairRootGroup/FairSoft.git fairsoft

cd fairsoft

git checkout feb26p1
```

Apply patch to correct some issues:

```
wget https://bmn.jinr.ru/plug/fairsoft_feb26p1.patch

patch -p1 -i fairsoft_feb26p1.patch
```

Installing FairSoft (-j option is used to parallel with multi-cores):

```
cmake -B build -DCMAKE_INSTALL_PREFIX=install DCMAKE_BUILD_TYPE=RelWithDebInfo -DGEANT4MT=OFF

cmake --build build -j4
```

3.Install the FairRoot environment
The release of FairRoot (18.8.3) can be downloaded from GitHub:

```
cd $INSTALLATION_PATH

export SIMPATH=$INSTALLATION_PATH/fairsoft/install

export PATH=$SIMPATH/bin:$PATH

git clone https://github.com/FairRootGroup/FairRoot.git fairroot

cd fairroot
```

```
git checkout v18.8.3
```

Apply patch to correct some issues:

```
wget https://bmn.jinr.ru/plug/fairroot_18_83.patch
patch -p1 -i fairroot_18_83.patch
```

Installing FairRoot (-j is used to parallel with multicores):

```
mkdir build

cd build

cmake -DCMAKE_INSTALL_PREFIX="$INSTALLATION_PATH/fairroot/install" -DBUILD_EXAMPLES=OFF ..
make -j4
make install
```

Installing the srcroot framework

- **Go to the installation directory**

NOTE: If you are installing srcroot to a non-system directory like home directory, then **do not use 'root' superuser.**
Go to the directory where you want to install the srcroot software, e. g. home directory:

```
cd ~
```

- **Clone GIT repository**

•Clone GIT repository of the last developer srcroot version with HTTPS (read-only access, e.g. for unregistered users)

```
git clone -b main --recursive https://github.com/tatovullaev-lgtm/srcroot
```

- **Installation procedure for srcroot:**

•Set environment:
NOTE: By default, in the SetEnv.sh file SIMPATH points to /opt/fairsoft/install, and FAIRROOTPATH - /opt/fairroot/install directories. If you installed FairSoft or FairRoot to another directory, please, change SIMPATH and FAIRROOTPATH variables in the file ("export SIMPATH=[your FairSoft install path]" and "export FAIRROOTPATH=[your FairRoot install path]").

```
cd srcroot

mkdir build

. SetEnv.sh
```

•Build the framework:

```
cd build

cmake ..

make -j4

. config.sh
```

Put relevant path for srcroot and remove string module add CMake/v3.26.5 libexpat libxml2 GSL in start_srcroot_cluster.sh

Running the srcroot

•Run “srcroot/start_srcroot_cluster.sh” every time you run terminal to work with srcroot